Classification of Cloud Phases in Cold-Air Outbreak Events Based on Polarization Lidar and Cloud Radar Observations

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***Abstract*—Cold-air outbreak (CAO) clouds significantly impact regional weather, precipitation, and aviation safety. This impact is especially pronounced in confluence zones, where cold continental air masses interact with warm, moisture-laden air currents emanating from oceans or lakes. In this article, an objective classification algorithm of cloud phase was suggested using the K-means clustering algorithm based on Polly^{XT} lidar method and a gradient-based cloud identification algorithm, in order to determine the depolarization ratio threshold for supercooled water, mixed-phase, and ice clouds. The analyses of cloud phase characteristics and vertical structure were conducted for CAO events over the Shandong Peninsula, a region frequently affected by sea-effect snow disasters, using the winter period of 2024–2025 continuous observation data from the Eastern China Cold-Air Outbreak Snowfall (ECHOES) campaign. The results show the following. The supercooled water clouds were predominantly located above the ice clouds in the mid-to-upper troposphere, implying heterogeneous ice nucleation processes. The analysis of diurnal variation reveals that the probability of occurrence of supercooled water clouds remains constant during the whole day because of the stable atmospheric conditions. Synergy of lidar and radar observations clearly demonstrates their complementary detection abilities: radar outperforming lidar in lower troposphere and daytime conditions. The high coverage of the supercooled water in the lower troposphere poses a significant safety threat for aircraft icing. These results provide an observational basis for improving the cloud microphysics parametric schemes in numerical weather prediction and aviation safety.**

***Index Terms*—Cloud phase classification, cloud radar, cold-air outbreak (CAO), ice cloud, lidar, supercooled liquid water (SLW).**

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I. Introduction

COLD-AIR outbreak (CAO) clouds, also known as lake-effect clouds or sea-effect clouds, are convective clouds with the potential for heavy snowfall when cold continental air masses traverse relatively warm water bodies [1], [2], [3]. The CAO process involves complex microphysical mechanisms, which include an increase in heat and moisture fluxes at the surface, destabilization of the atmospheric boundary layer, and the formation of organized convective cloud structures [4]. The significance of the CAO clouds extends beyond regional weather impacts, as these events contribute significantly to global climate variability, hydrological cycles, and pose severe socioeconomic threats through transportation disruptions, infrastructure damage, and agricultural losses [5], [6], [7]. In spite of their importance, CAO cloud events remain insufficiently understood due to their highly localized nature, rapid temporal evolution, and the challenges associated with observing cloud microphysical processes in adverse winter conditions [8]. Previous studies have centered on the climatology, decadal variations, and mechanisms of CAOs in key regions, the associated atmospheric boundary-layer processes, cloud characteristics, and extreme events, while also exploring the representation of related processes in numerical models and their impacts on regional weather and climate [9], [10], [11]. However, significant gaps in knowledge still remain when it comes to the detailed vertical structure of hydrometeor phase distribution, the changes in supercooled liquid water (SLW) content, and the ice nucleation processes in CAO clouds [12], [13]. The lack of observational studies, especially using multiplatform remote sensing data, inhibits the ability to validate numerical weather prediction models for this high-impact weather phenomenon [14].

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Conventional observational approaches for investigating CAO clouds have primarily relied on radiosonde soundings and aircraft-borne in situ measurements, which provide high-precision data for targeted locations but have constraints in covering large areas, making it challenging to conduct full-scale observations of CAO cloud systems [15], [16]. Satellite observations have revealed the large spatial extent and temporally evolving nature of CAO cloud systems [17]. However, the relatively low temporal cadence and the poor vertical detection capability limit the detail with which cloud microphysical properties can be retrieved [18].

Since the 1960s, with the development of ground-based remote sensing techniques, most notably light detection and ranging (lidar), a complementary tool for continuous monitoring of cloud structure and SLW has emerged [19]. Lidar is particularly useful in generating high vertical resolution data of cloud tops and bases, optical depths, and particle phase information through depolarization measurements. The capability of lidar to detect thin SLW layers, which play a crucial role in ice generation processes and precipitation development, has proven indispensable for advancing our understanding of mixed-phase cloud microphysics [20], [21]. Recent advances in lidar techniques have further enhanced the ability to distinguish between supercooled water, ice crystals, and mixed-phase clouds within complex cloud structures [22]. By determining the depolarization ratio, lidar capable of measuring the depolarization is useful for the phase classification of cloud particles, separating spherical liquid from nonspherical ice habitus [23]. Recent progress made in cloud phase discrimination has centered on optimizing depolarization thresholds designed for mixed-phase clouds, with the conclusion that SLW layers are present in clouds at the tops and often frequently occur at cloud tops containing ice precipitation [24].

Millimeter-wavelength cloud radar (MMCR) is crucial for cloud microphysics. High-resolution reflectivity and Doppler velocity profiles capture signatures such as Kelvin–Helmholtz instability and quantify ice particle growth in mixed-phase clouds—using dual-polarization parameters to track ice crystal evolution with SLW, linking dynamics to microphysics [25], [26], [27]. Its collaboration with lidar reveals critical coupled details of cloud vertical structure and microphysical composition, which lidar struggles to detect, together uncovering the full vertical distribution of cloud particles from thin upper layers to precipitating lower regions [28], [29], [30]. For CAO clouds, it determines ice and liquid water distributions, solves CAO microphysical uncertainties through secondary ice observations, and verifies model microphysical parameterizations, enhancing CAO comprehension and weather forecasting [31], [32].

Even though lidar and cloud radar systems each have their own strength, significant limitations remain when these devices are used alone for CAO clouds investigations [33]. Lidar signals experience strong attenuation in the presence of dense precipitation, often preventing detection of cloud layers above the first optically thick layer. In contrast, cloud radars exhibit reduced sensitivity to small cloud droplets and thin liquid layers that are critical for understanding ice nucleation processes [34]. The challenge of retrieving SLW content and distinguishing between different ice crystal habits using single-instrument approaches accurately has motivated the development of integrated retrieval algorithms that integrate lidar and radar observations [35], [36]. Shupe et al. [24] developed a ground-based multisensor system combining lidar and radar for robust phase classification, and Bühl et al. [37] showed that it is possible to detect thin supercooled water layers within mixed-phase systems. The integration of lidar with complementary instruments, like cloud radar and radiometers, is to describe hard-to-solve cloud structures and microphysics [38].

However, coordinated observation equipment and the relevant retrieval methods are still in an early stage of research; there have been very few comprehensive field campaigns that use coordinated lidar–radar observations to target specific CAO cloud events over eastern China. This observational gap has impeded the creation of region-specific comprehension of cloud phase distribution, precipitation efficiency, and the relative significance of different microphysical processes to CAO cloud production. Thus, we performed a continuous observation campaign in winter and spring on the Shandong Peninsula, next to the Bohai Sea, to investigate the detailed characteristics of CAO cloud events by combined lidar and cloud radar measurements [39].

This article is organized as follows. Section II presents the experimental setup, instrument properties, and data analysis technique utilized in this research. Section III gives the hydrometeor phase classified results of distinct cloud types observed in the campaigns. Section IV draws the conclusion; it remarks and discusses implications for future research directions.

II. Data and Methodology

A. Data

The Eastern China Cold-Air Outbreak Snowfall (ECHOES) campaign, conducted from November 1, 2024, to March 11, 2025, was jointly organized by Chinese Academy of Meteorological Sciences along with several other institutions, and 114 valid observation days were retained after excluding data affected by instrument malfunctions, signal interference, or other quality issues. Centered at the Yantai CAO Snow Observation Super Station, this experiment relied on Ka-band cloud radar and lidar as key equipment; other supplementary devices comprised of the C-band ultrahigh-resolution profile detection radar and G-band cloud radar. These instruments enabled refined observations of CAO snowstorms as well as individual snow crystals and have provided remote sensing of snowflake size distribution, density, type, concentration, and shape.

The ECHOES campaign was launched at Yantai Meteorological Bureau located in Shandong province, China. The geographic coordinates are: longitude 121.43°E; latitude 37.48°N, and 47 m in altitude. Adjacent to the Bohai Sea, this region experiences frequent CAO snowfalls. Weihai experienced a CAO from 10 to December 20, 2023, with a record-breaking snowfall of 115 mm. This confirms Weihai's significance as a key reference for our CAO cloud microphysics study on the peninsula [40]. In this study, all the time

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TABLE I
CLOUD PHASE CLASSIFICATION CRITERIA

Cloud Type	Temperature (°C)	Depolarization Ratio (δ)	Attenuated Backscatter (Mm^{-1} sr^{-1})	Description
Warm Cloud	$T > 0$	$0.069 \leqslant \delta < 0.138$	> 1	Pure liquid water droplets
Supercooled Water	$-38 < T \leqslant 0$	$0.069 \leqslant \delta < 0.138$	> 1	Liquid water below 0°C
Mixed-Phase	$-38 < T \leqslant 0$	$0.138 \leqslant \delta < 0.167$	> 1	Liquid + ice coexistence
Ice Cloud	$T \leqslant 0$	$\geqslant 0.167$	> 1	Pure ice crystals

records used Beijing Time (UTC+ 8) to ensure the uniformity and validity of the time data.

The main device of the experiment was Ka-band cloud radar [41]. The frequency of the Ka-band is 34.96 GHz, and it can achieve a sensitivity level lower than −30 dBZ. The detection height can be from the ground up to 15.3 km. The radar is particularly adept at accurately measuring echo intensities within the −45- to 30-dBZ range. Also, with a spatial resolution of around 30 m and a temporal resolution of 2 s [42], [43].

The Water vapor, Cloud, and Aerosol Lidar (WACAL) is a portable sensing device that employs the principle of Raman and elastic scattering for complete atmospheric characterization. Designed specifically to study the distribution of water vapor, cloud microphysics, and aerosols, it captures elastic backscattering signals at 355, 532, and 1064 nm. Notably, the 532-nm elastic scattering signal is used in this study. Such observational results support vertical profiling of cloud vertical structures, enabling the high-precision retrieval of multidimensional parameters, including water vapor mixing ratios and cloud base heights. It has a spatial resolution of 15 m, a temporal resolution of 1 min, and a maximum detectable height of 44 880 m. With its robust design, WACAL guarantees continuous and reliable observational capabilities under a broad range of environmental scenarios [44].

The European Centre for Medium-Range Weather Forecasts (ECMWF) Reanalysis v5 (ERA5) dataset is produced by the ECMWF and is a widely used global atmospheric reanalysis dataset [45], [46]. In this article, the data employed are hourly at 37 pressure levels in ERA5. The selected space point is longitude: 121.43°E and latitude: 37.48°N. The main variables include geopotential and temperature. Geopotential height was converted into a geometric height above the lidar station according to the standard conversion, $H = z/(g \times 1000)$, where z is geopotential and g is gravitational acceleration. Temperature values were interpolated from ERA5 pressure levels onto the lidar's vertical grid using linear interpolation. For the CAO identification part, the selected spatial domain covers the Shandong Peninsula region (37.5°N–38.5°N and 120.3°E–121.05°E). The key variables include sea surface temperature (SST), mean sea level pressure (SLP), as well as the atmospheric temperature and zonal (U) and meridional (V) wind components at 850 hPa. All ERA5 data were converted to local Beijing time (ERA5 UTC time + 8 h).

B. Methodology

CAOs were identified using a combined thermodynamic and dynamic criterion based on ERA5 reanalysis data following Bart Geerts' method [11]. The thermodynamic criterion utilizes the potential temperature difference (M) between the sea surface and 850-hPa level, defined as $M = \theta_{SST} - \theta_{850}$, where θ_{SST} and θ_{850} are the potential temperatures calculated from temperature and pressure at the sea surface and 850 hPa, respectively. The dynamic criterion requires the 850-hPa wind direction, indicating southwesterly to northwesterly flow. For each hourly time step, all variables were spatially averaged over the Shandong Peninsula region before applying the criteria. The CAO event was identified when both M >0 and the wind direction lies within 240°–360° based on the climatic characteristics of CAOs over the Shandong Peninsula. Using this methodology, a total of 1418 CAO hours were detected during the study period (see Table I of Supplementary Material).

For the cloud radar, quality control is applied to the Doppler spectrum during the observation period in order to accurately identify cloud and precipitation signals. Following the method proposed by Williams et al. [47], the influence of ground clutter on low-level clouds is eliminated. Then, based on the criterion that the number of consecutive valid spectral points exceeds 5, the range of the cloud signal is determined, and the cloud reflectivity factor is recalculated through integration [48]. The radar underwent absolute calibration upon initial installation to ensure the accuracy of echo intensity.

We processed raw photon count profiles acquired using the Polly^{XT} lidar methodology [49], following standard procedures to derive attenuated backscatter coefficients and volume depolarization ratios (see Fig. 1) [50]. Background noise was first estimated from the signal in the uppermost 5 km of each profile and subtracted from the raw photon counts. The normalized range-corrected signal (NRB), attenuated backscatter coefficient (β'), was subsequently derived by dividing the NRB by the lidar calibration constant, which was determined using the Rayleigh calibration method in cloud-free atmospheric regions. Volume depolarization ratios (δ) were computed from the perpendicular (S-channel) and parallel (P-channel) polarization components after applying dead-time correction and overlap function correction [51], [52]. The depolarization ratio was calculated as δ = S-channel/P-channel.

We used a gradient-based detection algorithm applied to the attenuated backscatter profiles to automatically identify cloud layers [53]. The algorithm employed a multistep procedure: 1) vertical smoothing through three point moving average to cut down noise but keep cloud edges; 2) calculating the first derivative of the smoothed backscatter profile to recognize sharp changes; 3) establishing cloud bottom heights where the vertical slope went past a 2–3 standard deviations threshold above general variability; and 4) recognizing cloud top heights

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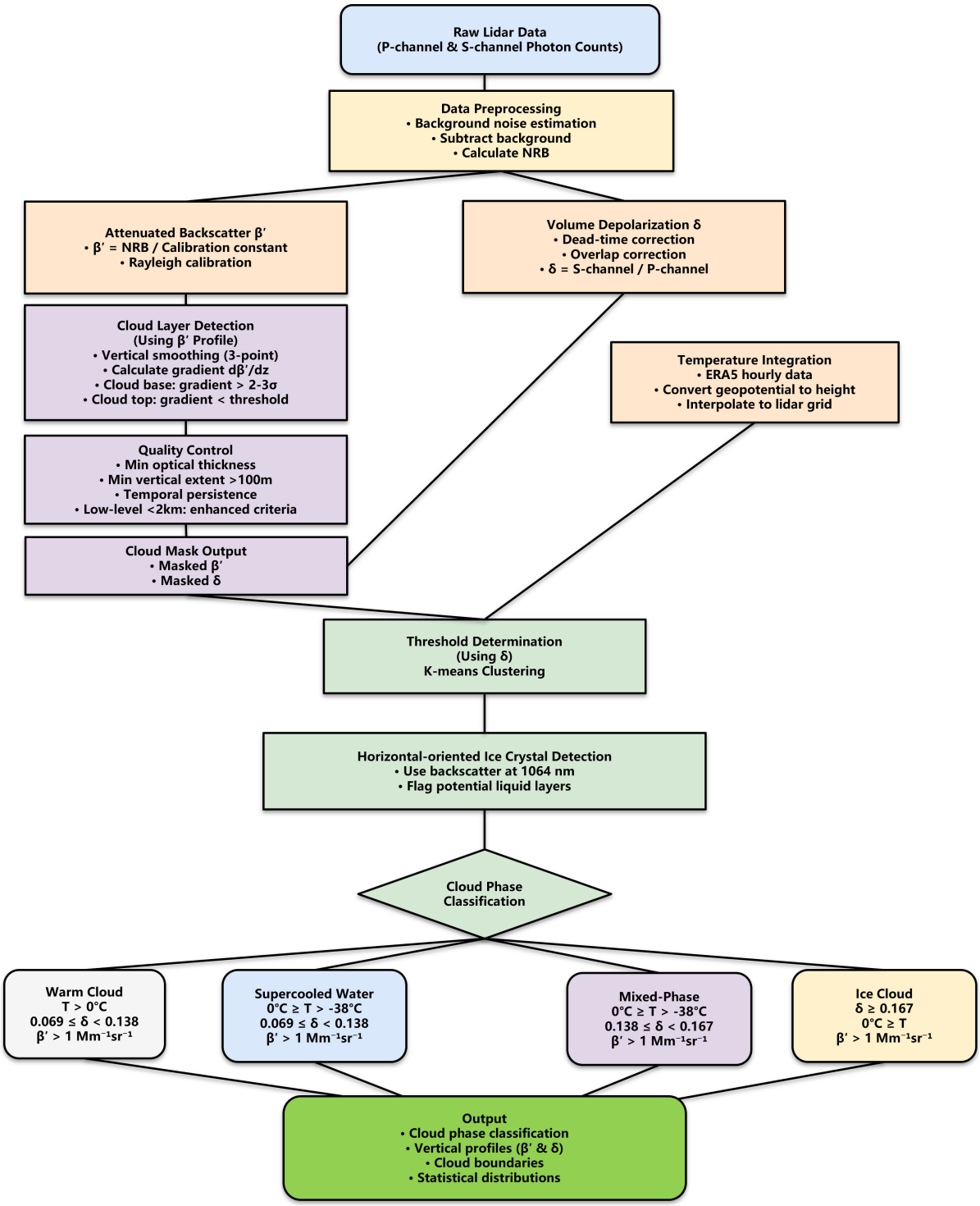


Fig. 1. Cloud phase classification method flowchart.

where the slope became negative and under the threshold. To eliminate false detections from aerosol layers and noise spikes, additional quality controls were applied, including minimum cloud optical thickness requirement, minimum cloud vertical extent constraints, and temporal check of consecutively having cloud profiles. For the low-level clouds below 2 km, enhanced detection criteria were implemented based on median backscatter intensity and 90th percentile thresholds to improve water cloud identification [54]. The algorithm output included cloud mask, masked attenuated backscatter, and masked depolarization ratio.

We analyzed the relationship between attenuated backscatter and depolarization ratio across different temperature regimes by combining lidar-retrieved cloud properties with hourly ERA5 temperature profiles. Fig. 2(a)–(f) shows the probability density function distributions stratified by temperature.

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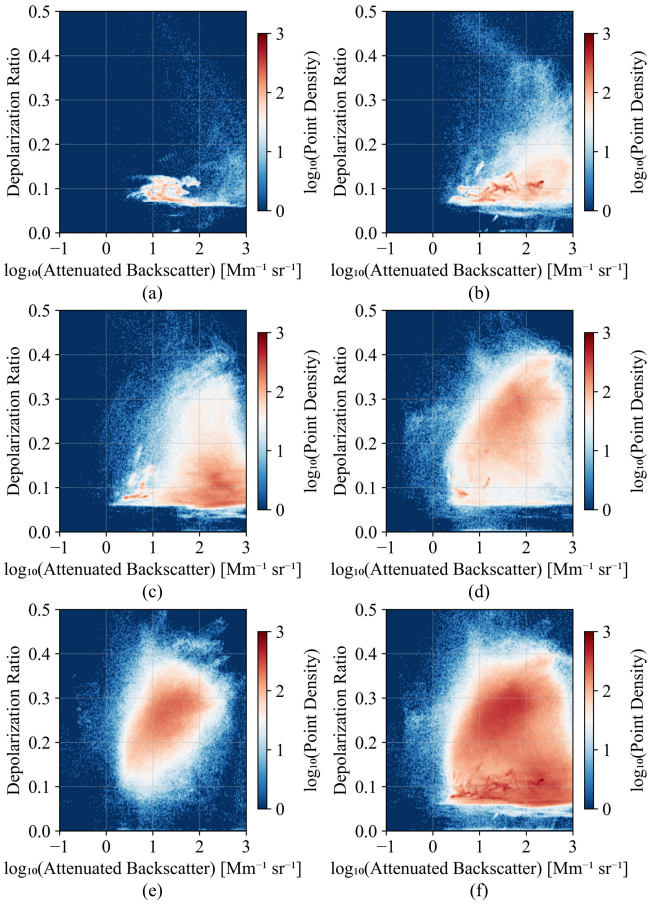


Fig. 2. Temperature-based cloud classification showing the relationship between attenuated backscatter coefficient and depolarization ratio. (a) Warm clouds (T > 0 °C and n = 71 325). (b) Mixed-phase region (0 °C ≥ T > −10 °C and n = 372 720). (c) Mixed-phase region (−10 °C ≥ T > −20 °C and n = 763 957). (d) Mixed-phase region (−20 °C ≥ T > −38 °C and n = 903 107). (e) Ice clouds (T ≤ −38 °C and n = 741 901). (f) All data combined (n = 2 853 010). The color scale represents log10 of point density.

Warm clouds [T > 0 °C, Fig. 2(a)] exhibit predominantly low-depolarization ratios characteristic of liquid droplets [see Fig. 2(a)]. Mixed-phase clouds appear in the ranges of 0 °C to −38 °C [see Fig. 2(b)–(d)]. Following Rosenfeld and Woodley [55], who found that most condensed water remains liquid down to −37.5 °C, we selected −38 °C as the threshold for pure ice phase [see Fig. 2(e)]. We can see from Fig. 2 that as the temperature decreases, cloud particles tend to cluster toward higher depolarization ratios.

To distinguish horizontally oriented ice crystals from SLW clouds, we utilized a backscatter-based detection algorithm derived from the Polly^{XT} lidar analysis framework, which exploits the characteristic strong attenuation of laser signals by liquid water droplets [50]. The algorithm uses the particle backscatter at 1064 nm, identifying cloud layers where the signal shows rapid decay over a fixed vertical distance, indicative of liquid water presence. For each detected attenuation signature, the cloud base and top are identified through gradient analysis within predefined search windows, while low-depolarization layers without attenuation signatures are classified as horizontally oriented ice crystals.

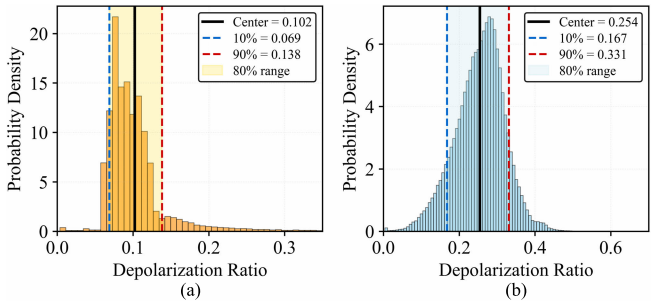


Fig. 3. K-means threshold classification based on depolarization ratio. (a) Warm clouds (T > 0 °C) with center at 0.102 (black solid line), 10th percentile at 0.069 (blue dashed line), and 90th percentile at 0.138 (red dashed line). (b) Ice clouds (T < −38 °C) with center at 0.254 (black solid line), 10th percentile at 0.167 (blue dashed line), and 90th percentile at 0.331 (red dashed line). The shaded areas represent 80% ranges.

To objectively determine the depolarization ratio thresholds for distinguishing water and ice phases, we employed K-means clustering with the 80th percentile method on warm cloud (T > 0 °C) and ice cloud (T < −38 °C) data [see Fig. 3(a) and (b)]. The water cloud center was identified at 0.102, and the ice cloud center was identified at 0.254. For warm clouds (a representative of water-dominated phases), the depolarization ratio range was constrained by the 10th and 90th percentiles of the warm cloud dataset, corresponding to 0.069 and 0.138, respectively—thus defining the water phase threshold as $0.069 \leq \delta < 0.138$. For ice clouds, the 10th percentile of the ice cloud dataset was determined as 0.167, which was set as the lower threshold for pure ice phases ($\delta \geq 0.167$). The interval between 0.138 (upper bound of water phase) and 0.167 (lower bound of ice phase) was therefore designated as the transition range for mixed-phase clouds. This approach includes 80% of all observations for each cloud type (the 10th–90th percentile of depolarization ratio for both warm and ice clouds), allowing for reliable SLW detection in mixed-phase clouds. In the transition region of $0.138 < \delta < 0.167$ between water and ice thresholds, clouds are deemed to be mixed phase when the attenuated backscatter coefficient is above 1 Mm^{-1} sr^{-1} [see Fig. 2], which indicates sufficient signal quality for reliable phase discrimination. In detail, the different cloud phase is classified according to specific temperature, depolarization ratio, and attenuated backscatter coefficient ranges as shown in Table I.

III. Results

A. Cloud Phases' Vertical and Thermal Distribution

The height–temperature distribution of supercooled water clouds and ice clouds was analyzed during the CAOs using lidar observation data [see Fig. 4]. The pdf derived from this dataset gives the occurrence frequency of every cloud phase over different height–temperature bins, which indicates different vertical stratification and thermal dependency for SLW clouds and ice clouds.

Supercooled water clouds [see Fig. 4(a)] mostly appear in the lower troposphere (1–2 km) with peak occurrences between −5 °C and −20 °C. Notably, the pdf of SLW clouds drops off steeply when temperatures drop below −20 °C.

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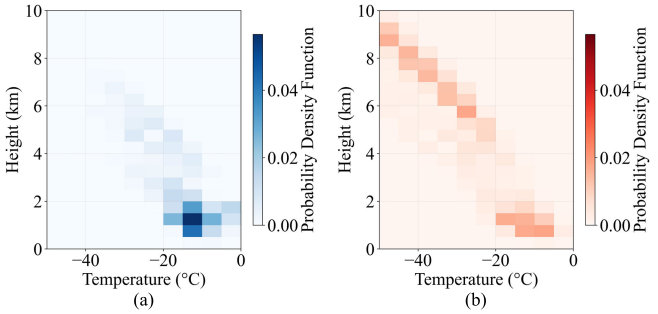


Fig. 4. Height–temperature distribution during CAO events. (a) Height–temperature distribution of supercooled water clouds. (b) Height–temperature distribution of ice clouds. The probability density function (pdf) depicts the occurrence frequency of clouds at different height and temperature combinations.

This sharp decline is likely driven by the temperature-dependent activation efficiency of ice nucleating particles (INPs); −20 °C is above the homogeneous freezing threshold of cloud droplets (−38 °C), so the kinetic energy of water molecules is insufficient to trigger autonomous freezing. Instead, the subfreezing temperature regime below −20 °C enhances INP activation efficiency, with Asian dust, the most widespread INP in East Asia, acting as a key contributor via immersion freezing mode, thereby promoting heterogeneous nucleation and accelerating the conversion of SLW to ice [56], [57].

On the other side, ice clouds have a wide vertical spread and thermodynamic spread (from bottom of troposphere to top of troposphere) covering the entire column of atmosphere, where the most probable density is present near 5–9 km in height and −25 ° to −45 °C. The ice clouds were found around 5–9 km, the place where the probability density gets its highest value, which is −25 ° to −45 °C. This upper tropospheric region is characterized by cold temperatures that favor efficient ice crystal growth and low INP concentration, enabling ice crystals to persist and aggregate into large-scale cirrus or cirrostratus layers.

B. Diurnal and Vertical Distribution of Different Phase Clouds

We derive the diurnal height distribution of cloud occurrence frequency by binning the lidar cloud observations during CAO events, where we classify cloud phases based on the depolarization ratio thresholds and attenuated backscatter threshold [see Table I].

The vertical distributions display physically consistent patterns according to the thermodynamic constraint. Supercooled water clouds are mainly located below 2 km, and the peak frequency is 20.88% at a peak height of 1.25 km, which is consistent with temperature-driven phase transitions [see Fig. 5(b)]. Above 3 km, where temperatures are usually lower than −20 °C, ice nucleation becomes the main process, and the frequency of supercooled water decreases rapidly. Ice clouds show a bimodal structure; the lower level has a maximum below 2 km for precipitating ice and the upper one is at 4–10 km, typical for cirrus [see Fig. 5(d)]. Total cloud frequency exhibits a pronounced maximum in the boundary layer

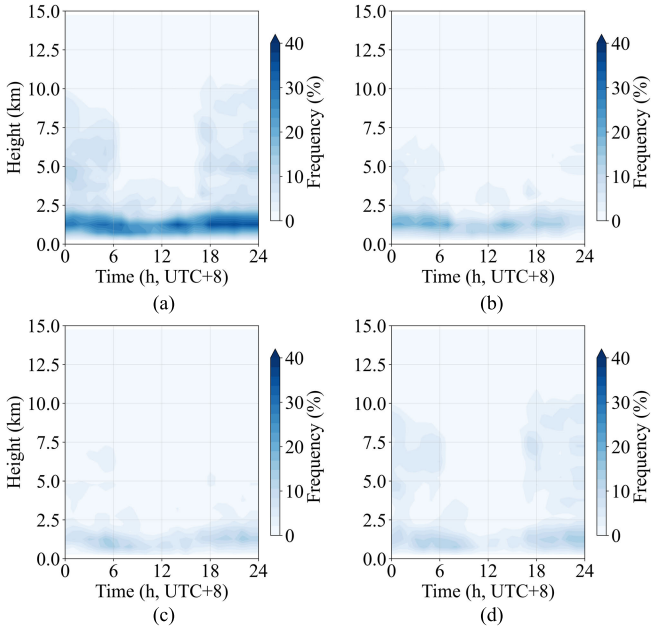


Fig. 5. Diurnal height distribution of cloud occurrence frequency during CAO events. (a) Total cloud frequency. (b) Supercooled water cloud frequency. (c) Mixed-phase cloud frequency. (d) Ice cloud frequency. Frequency indicates the percentage of time that clouds were observed at different height–time combinations.

(below 2 km, reaching 36% at 1.25 km), with significantly lower frequencies in the upper troposphere (8–10 km, 2%–6%) [see Fig. 5(a)].

Supercooled water clouds [see Fig. 5(b)] show minimal diurnal variation, maintaining basically constant frequency over a 24-h period. This weak diurnal signal suggests that control on the boundary-layer liquid clouds is by a more stable synoptic forcing, rather than local solar radiative forcing. Contrary to this, it was found that there is an apparent high strong diurnal change of total cloud frequency [see Fig. 5(a)], especially the decrease at daytime 06:00–18:00, which is partially because of the lidar's instrumental limitation. During the day, more solar background photons make it harder to see things in the clouds because they make the signal-to-noise ratio worse. In the following, this systematic bias will be quantitatively assessed by comparing it to Ka-band radar observations.

Further support for the presence of a daytime diurnal pattern of cloud vertical distribution is given by the Ka-band radar observations [see Fig. 6(a)]. Specifically, in the 10:00–18:00 period, the radar observes an obvious increase in the height of the main cloud frequency center, that is, the area of high-frequency clouds expands upward to 6–9 km. The upward movement of clouds is mainly caused by solar radiative heating of the surface. From 10:00 to 18:00, intense solar shortwave radiation absorbed by the surface (land and Bohai Sea) of the Shandong Peninsula raises surface and boundary-layer temperatures. The enhanced surface heating sets up a more steeply sloped vertical temperature gradient that destabilizes the lower troposphere and enhances convective mixing—weak updrafts (consistent with CAO dynamics) transport moisture and cloud droplets upward from the surface to higher levels in the atmosphere, which lifts the whole

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cloud base and increases the height range where clouds occur vertically. The solar-driven cloud elevation is different from the lidar-observed daytime frequency drop (attributed to instrumental bias) and represents a thermodynamic-dynamical response of CAO cloud systems to diurnal solar forcing.

Supercooled water dominating the lower part of the troposphere indicates that the persistence of liquid water at subfreezing temperatures is a key factor in facilitating the formation and sustenance of CAO precipitation. Transitioning from supercooled water to ice with increasing height represents the temperature dependence of ice nucleation efficiency, with mixed-phase clouds representing the 2–5-km transitional layer [see Fig. 5(c)].

SLW cloud and ice cloud complementary vertical distribution shows that there is a clear stratified distribution, that is, SLW clouds are mostly distributed in the lower troposphere (1–2 km), and ice clouds are mostly distributed in the middle and upper troposphere (4–10 km). Between the two regions lies a distinct mixed-phase transition zone, which is specifically located at altitudes of 2–4 km and within the temperature interval of −10 °C to −20 °C; SLW and ice crystals coexist in this zone. This zone is dynamic and cloud microphysics properties heavily influence the occurrence of CAO precipitation; the presence of SLW provides water vapor for the growth of ice crystals via the Wegener–Bergeron–Findeisen (WBF), and ice crystals act as nuclei for additional precipitation [58].

C. Comparison of Cloud Detection Between Lidar and Radar

The radar observations show that the clouds are most frequent below 2 km and approximately reach 50% [see Fig. 6(a) and (b)]. There is a secondary maximum observed around 6–9 km for nighttime (18:00–06:00 local time) cloud frequencies are observed around 10%–15%. The vertical shows a decreasing mean cloud frequency as a function of height, with the largest variability (min–max range) occurring in the lower troposphere below 2 km [see Fig. 6(b)]. To study whether there is any attenuation effect on the lidar cloud detection, we compared the cloud frequencies detected by radar and lidar by computing the difference (lidar minus radar) [see Figs. 5(a), 6(a), and (c)]. The comparison shows that there are three characteristics of the lidar performance under different atmospheric conditions.

First, significant negative differences are observed below 2 km for all of the diurnal cycle [see Fig. 6(c)], with a mean difference near the surface being around −30% [see Fig. 6(d)]. This substantial underestimation of the lidar is mainly because there is heavy snowfall, and it attenuates the signal strongly. During precipitation, the lidar beam is strongly attenuated by snow particles in the lower atmosphere, resulting in inadequate detection of cloud layers above. On the other hand, the Ka-band radar that works at 35 GHz is better to penetrate the precipitation and can detect cloud in the lower part of troposphere more reliable.

Second, pronounced negative biases are observed for all the altitude ranges during daytime (06:00–18:00 local time), which is a systematic bias due to a fundamental limitation of lidar measurements. During the daytime, increased solar

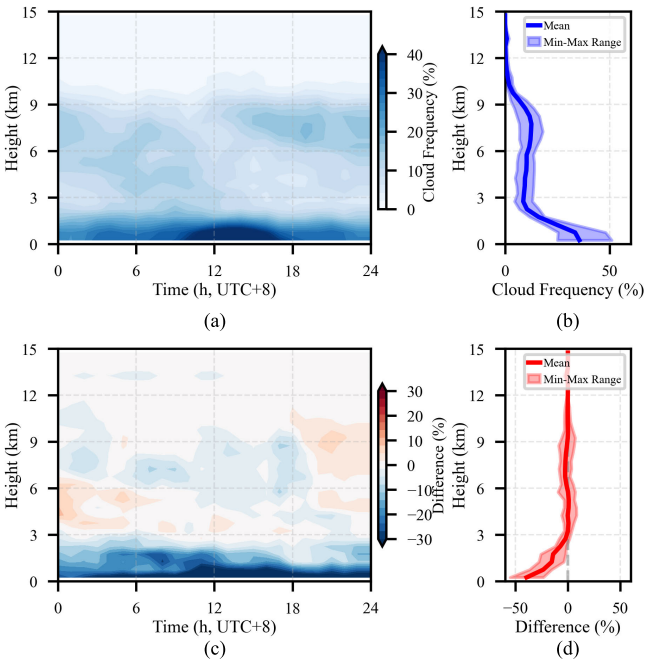


Fig. 6. Cloud frequency diurnal variation and observational comparisons between lidar and Ka-band radar during CAO events. (a) Time and height distribution of Ka-band radar cloud frequency, presenting the complete 24-h diurnal cycle. (b) Vertical profile of radar cloud frequency, with the blue solid line representing the mean frequency, while the shaded area indicates the min–max variation range during the diurnal cycle. (c) Diurnal variation of cloud frequency difference between lidar and radar observations (lidar minus radar). (d) Vertical profile of the frequency difference, with the solid red line showing the mean difference, while the shaded area representing the min–max range.

background radiation leads to a decline in the signal-to-noise ratio (SNR), and since the cloud detection algorithm depends on the SNR threshold, it lowers the sensitivity of the algorithm, leading to the positive difference in Fig. 6(c). This is an inherent limitation of lidar systems that are operating in the presence of sunlight.

Third, comparing lidar and radar at nighttime for upper level shows excellent agreement above 6 km as the difference is close to zero [see Fig. 6(c)]. In particular, lidar detects more frequent clouds than radar at 9–11 km during the night (18:00–06:00 local time), with positive differences up to 5%–7% [see Fig. 6(c)]. The improved detection performance of lidar at high altitudes at night is due to two reasons. 1. The lack of solar background noise creates ideal SNR conditions. 2. Lidar has a higher sensitivity to optically thin cirrus clouds that are below the radar detection limit. Vertical profile in Fig. 6(d) shows the mean difference transitioning from strongly negative close to the surface to near zero at higher levels and the minimum variability is found between 6 and 9 km, where the two instruments perform comparably.

These results show that radar performs more reliably for detecting clouds at low altitudes and during the day, while lidar offers superior sensitivity in detecting optically thin high-altitude clouds at night. The complementary features of these two remote sensing techniques emphasize the significance of synergistic observations for the comprehensive characterization of clouds.

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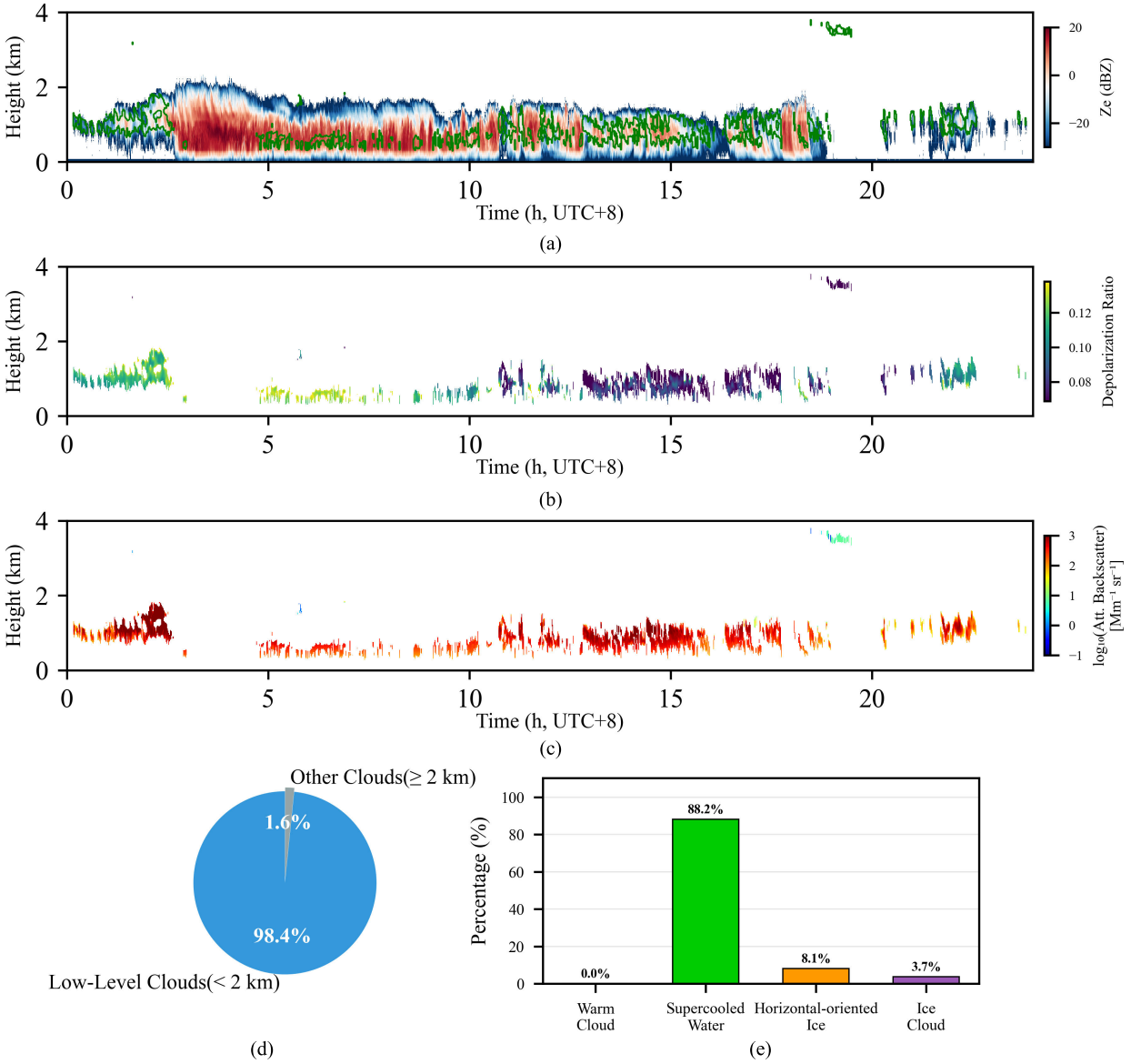


Fig. 7. Comprehensive radar-lidar observation on February 4, 2025, with supercooled water detection. (a) Ka-band reflectivity with supercooled water regions outlined in green. (b) Depolarization ratio for cloud regions only. (c) Attenuated backscatter coefficient ($\log_{10}$ scale) for cloud regions only. (d) Vertical distribution of clouds (low level <2 km versus higher clouds) during CAO events. (e) Phase composition of low-level clouds (warm, supercooled water, horizontal-oriented ice, and ice) during CAO events, based on cumulative statistics from the ECHOES campaign.

D. Supercooled Water Detection in Low-Level Precipitation Systems

During the ECHOES experiment, low-level precipitation systems that are associated with an abundance of SLW are frequent and climatologically significant over the Shandong Peninsula [see Fig. 7]. These systems shared three key characteristics that distinguish them from typical midlatitude winter precipitation clouds: shallow cloud structures, longer persistence in time, and pervasive SLW presence throughout the precipitation lifecycle, all of which were uniquely recorded by the cloud radar and polarization lidar system deployed in concert.

Almost all the low-level detection precipitation systems were confined to under 2 km high [see Fig. 7(a) and Supplementary Material Figs. 1 and 2]. Small vertical scale is an embodiment of thermodynamic structure in the thermodynamic structures that form CAO events; a dense and cold continent air mass acts as a "capping layer" by hindering the generation of deep convection and preventing large upward transport of moisture into the troposphere. Radar reflectivity observed that the snow systems were weak-to-moderate reflectivity echo (−25 to 10 dBZ) that is consistent with a light-to-moderate amount of snowfall, a feature of sea-effect precipitation driven by moisture from the adjacent Bohai Sea. Temporally, the SLW-bearing precipitation systems were characterized by considerable persistence, with a median duration of 11.0 h, a mean duration of 31.5 h, and a maximum duration of 155 h for individual events. [see Fig. 7(a)–(c) and Supplementary Material Table I and Figs. 1 and 2]. This timescale is much longer than that of typical convective (1–3 h) and is attributed to the steady-state moisture supply

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from the Bohai Sea. During the CAO, the continuous northerly wind and northeast wind carry a large amount of dry and cold-air southward [59]. When this cold air moves over the sea, it interacts with the moisture-laden maritime air there, thus promoting the sustained development of supercooled water, and the precipitation process can last for several hours.

The radar observations in Fig. 7(a)–(c) exposed several precipitation occurrences, green contours denoting supercooled water areas recognized by the lidar through the depolarization ratio criteria. Mostly located at less than 2 km and exhibited remarkable temporal continuity. The lidar-derived properties in supercooled water areas [see Fig. 7(b) and (c)] revealed that the depolarization values were all within the supercooled water range and there were strong backscattering signals. These observations collectively point to a substantial liquid water content. Though the lidar signal was severely attenuated in the most precipitating areas, the technique provided superior capability for detecting fine-scale supercooled water in regions of moderate reflectivity.

Cumulative statistics during CAO events to assess the frequency of low-level clouds and supercooled water at this site. As seen in Fig. 7(d), low-level clouds (cloud base <2 km) accounted for 98.4% of the total number of cloud detections. By identifying the phases of these low-level clouds, Fig. 7(e) reveals that a striking dominance of supercooled water (88.2%) over warm clouds (0.0%), horizontal-oriented ice (8.1%), and ice clouds (3.7%). The presence of supercooled water through these long events, even though there is ongoing ice precipitation, means that there must be a continuous supply of liquid water and that the ice nuclei are not very efficient. These results are consistent with CAO cloud properties, where the dominance of supercooled water over ice (98.4% versus 1.6%) indicates the scarcity of efficient INPs in these systems [60], [61].

There are two interconnected elements, both related to the specific maritime–continental interaction of CAO events over the Shandong Peninsula, which explain this unusual occurrence of SLW. First, the ongoing moisture supply from the Bohai Sea's warm water is the main cause for the creation of SLW, because it releases substantial latent heat and moisture fluxes into the upper layer of the overlying cold-air mass; when this moisture is quickly cooled to subfreezing temperature while being blown inland across the cold peninsula, it stays in its liquid form as SLW since there are no effective INPs that can efficiently trigger the freezing process, thereby creating a "moisture surplus" environment with abundant SLW content. Second, and more importantly, due to the shortage of efficient INPs in the CAO boundary layer above the Shandong Peninsula, there is a microphysical bottleneck in the cloud system. INPs are important for initiating the formation of ice crystals in subfreezing liquid clouds via heterogeneous nucleation, but the scarcity of INPs hinders the transformation of SLW to ice as ice particles fall through the layer as snow, unlike continental winter clouds. Therefore, the 88.2% SLW fraction of low-level clouds [see Fig. 7(e)] is not accidental but directly due to the moisture-INP imbalance, which is the unique fingerprint of CAO events over maritime-influenced mid-latitude coastal regions [62].

E. Supercooled Water Clouds Above Ice Clouds in the Mid-to-Upper Troposphere

A critical observation from the ECHOES campaign is the frequent vertical stratification of cloud phases in the mid-to-upper troposphere (altitudes >2 km) during the CAO events. SLW clouds consistently are located above ice clouds. This phase configuration, distinct from the well-mixed or ice-dominated structures of typical mid-latitude winter clouds, is not a random occurrence but a direct consequence of coupled thermodynamic, microphysical, and dynamical processes.

It can be seen from the comprehensive cloud classification analysis for the period of December 20, 2024 (00:00–12:00 BJT) that supercooled water clouds often exist above ice clouds in the mid-to-upper troposphere (above 2-km altitude). Cloud classification through time [see Fig. 8(a)] exhibits supercooled water (blue) frequently occurring above ice clouds (red) and mixed-phase clouds (green) being the transition region between those two distinct layers. The Ka-band radar reflectivity [see Fig. 8(b)] offers Supplementary Information on cloud vertical. The strong radar echoes in the lower ice cloud layers denote vigorous ice crystal growth and aggregation, whereas the weaker echoes in the upper supercooled water layers imply smaller droplets and fewer ice crystals. This vertical stratification is a defining feature of typical layered mixed-phase clouds. Primary ice nucleation starts at the liquid cloud top, where ice formation mainly occurs within supercooled liquid clouds [63]. Being relatively warm for immersion freezing, primary ice nucleation activity is weak, resulting in sparse initial ice crystals. These tiny crystals grow via the WBF process, settling from the shallow liquid layer to form ice virga without depleting the upper liquid phase. The limited ice crystal quantity, combined with cloud-top longwave radiative cooling that sustains condensation, enables the upper supercooled liquid layer to persist for a long time, consistent with long-lived supercooled layer cloud traits [64]. Meanwhile, vigorous updrafts continuously replenish moisture to the upper layers, promoting condensation and maintaining a liquid-dominant state at subfreezing temperatures. Conjointly, these processes support the stable "supercooled water clouds above ice clouds" vertical stratification.

Statistical analysis [see Fig. 8(e) and (f)] quantifies the cloud type and supercooled water clouds where ice clouds exist below: ice clouds dominate the mid-to-upper cloud population; however, supercooled water clouds have a high occurrence when ice clouds exist below (77.6%), versus 22.4% without ice clouds below. This implies that mid-to-upper cloud vertical phase structure is comprised of ice clouds at low cloud levels and supercooled water clouds above, due to multiple coupled processes, including: 1) vertical INP concentration gradients where boundary-layer transport, entrainment, or convergence zones deliver enhanced INP availability at lower levels; 2) liquid water depletion at lower levels via the WBF process as ice crystals grow at the expense of supercooled water droplets; 3) maintenance of supercooled water in the upper layer by shallow strong updrafts that efficiently transport water vapor upward from the lower troposphere, sustaining continuous condensation in the upper cloud layer; and 4) depletion of

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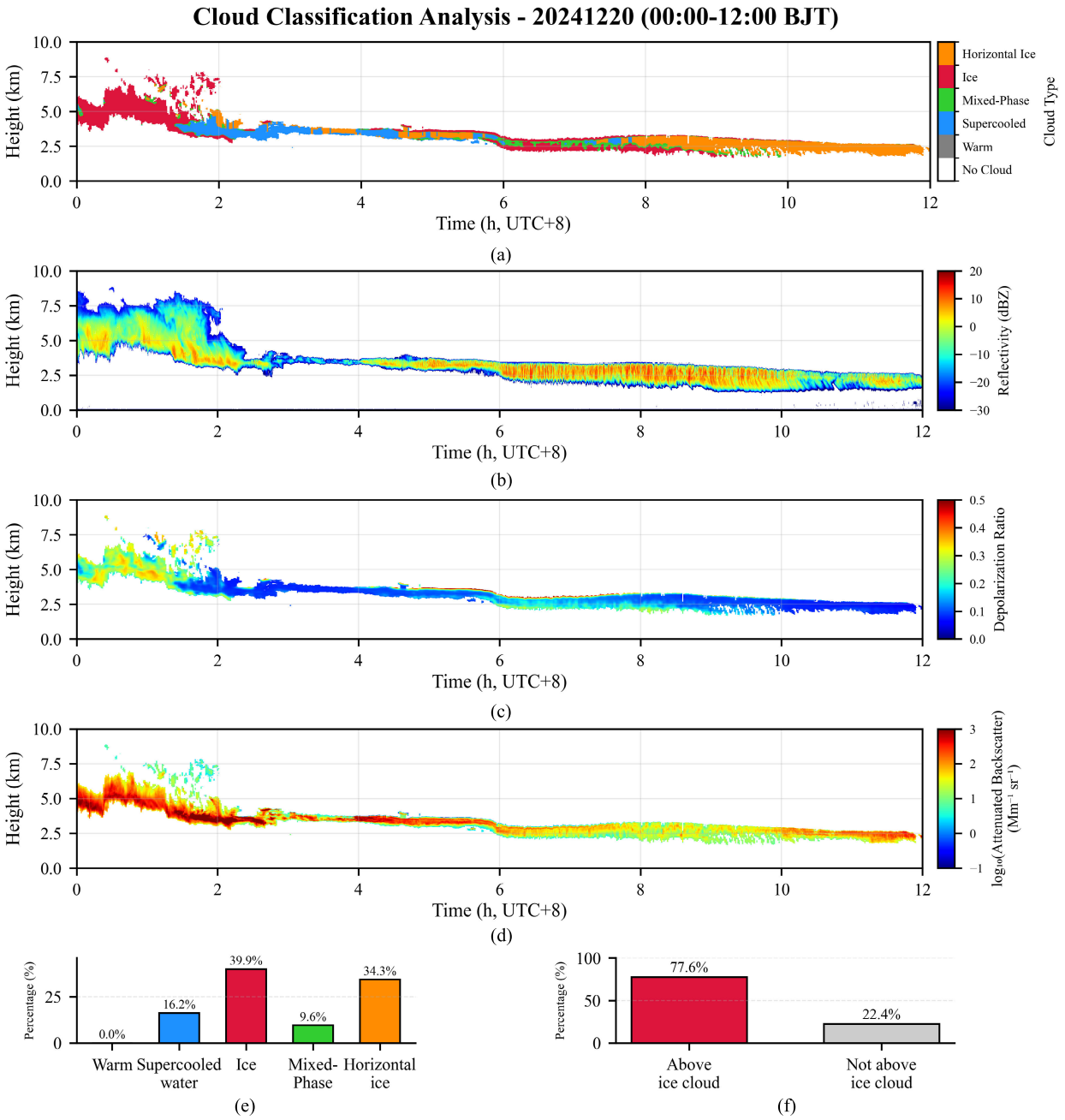


Fig. 8. Comprehensive cloud classification analysis for December 20, 2024 (00:00–12:00 BJT) during CAO periods. (a) Cloud classification showing warm (orange), supercooled water (blue), mixed-phase (green), and ice clouds (red). (b) Ka-band radar reflectivity showing cloud vertical structure. (c) Depolarization ratio distribution. (d) Attenuated backscatter coefficient ($\log_{10}$ scale). (e) Cloud type statistics during CAO periods, showing the percentage distribution of five cloud types: warm clouds, supercooled water, ice clouds, mixed phase, and horizontal-oriented ice. (f) Percentage of supercooled water clouds (above 2 km) with ice clouds or horizontal-oriented ice below versus without ice below, calculated during CAO periods only.

INPs in the upper layer caused by limited vertical transport of solid INP particles and consumption of INPs via ice nucleation in the lower ice cloud layer [65], [66]. These results improve our knowledge on cloud phase structures in the mid-to-upper troposphere, as well as the cloud–radiation interaction and the process of precipitation initiation through ice nucleation.

IV. Conclusion and Discussion

This study provides an overall view of cloud phase characteristics of CAO events over Shandong Peninsula through continuous observation from the ECHOES. Combining lidar and cloud radar observations, an objective cloud phase classification algorithm is developed to examine the vertical structure, temperature distribution, and temporal variations of SLW and ice clouds in CAO systems.

The cloud phase classification algorithm can successfully distinguish water, ice, and mixed-phase clouds based on the depolarization ratio derived from K-means clustering analysis. The water phase threshold and ice phase threshold were objectively determined by using the 80% percentile method. Using lidar and radar in synergy proved essential for comprehensive

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characterization. Lidar provided high-resolution detection of thin supercooled water layers through depolarization observations, while radar captured the precipitation structure and deeper cloud layers in which lidar signals were attenuated.

The vertical and thermal distribution analysis showed that different cloud phases have distinct characteristics. Supercooled water clouds are mainly concentrated in the lower troposphere (1–2 km) with peak occurrence at temperatures between −5 °C and −20 °C. The ice clouds exhibited a broader vertical distribution extending to 10-km altitude with maximum probability density at 5–9 km, and the maximum value is −25 °C to −45 °C. Based on the diurnal change analysis, the result shows the phase-dependent time, which has almost negligible diurnal variations in supercooled water cloud phases under the control of stable conditions, while ice cloud exhibited pronounced diurnal modulation influenced by the thermal forcing of solar radiation.

A striking finding was the prevalence of supercooled water in low-level precipitation clouds. Statistical analysis revealed that supercooled water dominated (88.2%) over warm clouds (0.0%), horizontal-oriented ice (8.1%), and ice clouds (3.7%) in low-level clouds below 2 km during CAOs. The persistence of supercooled water throughout CAO events, while ice precipitation is still taking place, indicates a continuous supply of liquid water and less efficient ice nucleation process, which is in line with a scarcity of efficient INPs in CAO systems. Also, supercooled water clouds were often found to be present above ice clouds in the mid-to-upper troposphere, which indicated that ice nucleation occurred heterogeneously and that INPs activated preferentially at lower cloud levels.

These findings have significant implications for our understanding of cloud microphysical processes and improving the understanding of numerical weather prediction. The observation that supercooled water dominates suggests that WBF is relatively inefficient at CAO, and thus, other secondary ice production mechanisms may play important roles [67], [68]. Vertical phase structures are a challenge to simplifying assumptions about cloud microphysics parameterizations and provide important constraints for improving the formulation of ice nuclei. Moreover, the supercooled water is more abundant below 2-km altitude with a total amount of 88.2% of the low-level clouds, which poses significant aviation safety hazards, as the icing will cause serious airplane icing when the aircraft flying through these areas [69]. These observations can provide critical information for aviation weather forecasting and flight safety management during the CAO events.

Our observation that supercooled water clouds occur at temperatures between −5 °C and −20 °C with peak occurrence in the 1–2-km altitude range is consistent with the COMBLE campaign results, which reported similar boundary-layer cloud structures during Arctic CAO events [11]. However, our findings reveal key regional differences specific to mid-latitude CAO systems. The persistence of supercooled water throughout precipitation events with a median duration of 11.0 h, a mean duration of 31.5 h, and a maximum duration of 155 h was longer than that reported in previous studies. For instance, collections in prior research were made over periods of 6–74 h (average 34 h) [70]. Despite ongoing ice precipitation, this extended persistence indicates a continuous moisture supply from the Bohai Sea, a characteristic that distinguishes these mid-latitude maritime CAO systems from continental winter clouds. Notably, the CAO clouds examined in this study are all below 2 km in altitude, marking a narrower research scope with more stringent constraints relative to the 4-km altitude range covered in prior work, a distinction that may be linked to regional characteristics and variations in convective intensity [71]. The scarcity of efficient INPs in the CAO boundary layer over the Shandong Peninsula maintains the observed 88.2% supercooled water fraction, representing a unique fingerprint of CAO events over maritime-influenced mid-latitude coastal regions. These findings provide new observational constraints on CAO cloud microphysics, demonstrating that while the fundamental thermodynamic processes are universal across CAO regions, the vertical phase structure exhibits significant regional variability controlled by geographical configuration and aerosol characteristics. The high aviation icing hazard posed by the extensive supercooled water presence below 2-km altitude represents a critical regional safety concern that warrants specific attention in mid-latitude CAO forecasting systems.

There are several limitations to this study. First, lidar derived diurnal cloud frequency pattern suffers from a systematic bias due to solar background radiation. During the daytime (06:00–18:00), increased solar background lowers the SNR, thus degrading the SNR thresholds for the cloud detection algorithm. This instrumental restriction causes a low estimation of daytime cloud occurrence frequency, which is shown by the large negative difference between lidar and radar observation during the sunlit time. Second, multiple scattering effects in optically thick water clouds can impact the accuracy of lidar depolarization measurements and cause overestimations of depolarization ratios in dense liquid layers [72]. This artifact is distinctly manifested in our observational data, as can be seen in Fig. 8(a). Specifically, the distribution of ice clouds above the supercooled water layer detected between 04:00 and 10:00 is inconsistent with fundamental physical principles governing cloud microphysical processes—supercooled water layers are thermodynamically unfavorable for the stable existence of ice phase particles in their upper regions under the observed ambient conditions. Such an unphysical observation can be predominantly attributed to the interference of multiple scattering effects. Corroborating evidence for this causal relationship is provided in Fig. 8(c), where a conspicuous abnormal increase in depolarization ratios is recorded during the same time window. The temporal synchronization between the spurious ice cloud signal and the anomalous depolarization enhancement strongly confirms that multiple scattering-induced distortion is a non-negligible factor in the interpretation of lidar depolarization data for optically thick cloud systems. Third, strong signal attenuation by lidar in heavy precipitation and snowfall limits observations to only thin surface layers, preventing complete vertical profiling during intense precipitation events [73]. Fourth, the horizontally oriented ice crystals can significantly reduce the lidar depolarization ratio, thereby causing misclassification of ice-containing cloud layers as supercooled water clouds [54].

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These limitations highlight the complementary value of synergistic lidar–radar observations. Radar gives a better detection of clouds in the lower troposphere, during daytime, and during precipitating conditions, but lidar has better sensitivity for optically thin high-altitude clouds during nighttime.

The future research can be carried out in the following directions. First, the classification algorithm could be improved by incorporating extra retrieval parameters, such as multiwavelength Raman scattering and radar Doppler spectra analysis for bimodal distribution detection. Second, the extensive supercooled water dataset provides opportunities to investigate ice nucleation mechanisms by analyzing relationships between supercooled water occurrence, temperature, and aerosol properties. Third, observations of vertical phase structures should be used to evaluate and improve parameterizations of cloud microphysics in numerical models, especially for INP availability and secondary ice production. Third, oblique lidar observations should be incorporated to detect horizontally oriented ice crystals, which can enhance the accuracy of supercooled water identification. Finally, quantitative microphysical retrievals that combine lidar and radar observations can be employed for the detailed studies of the precipitation within the CAO clouds.

This study shows the use of combined polarized lidar and cloud radar observations to characterize the cloud phase during CAO events. The observational dataset provides valuable insights into cloud microphysical processes and offers observational constraints for improving weather forecasts for high-impact CAO snowstorms. As observational networks expand and retrieval algorithms improve, such measurements will play an increasingly important role in advancing our understanding of cloud microphysics and validating numerical models.

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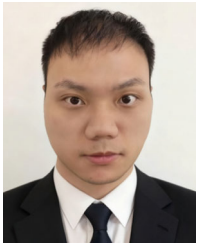
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